

resolved with each successive generation of 3D printers and is expected to match up to traditional manufacturing methods within the next ten years.

With investment in 3D printers for in-house use already seeing a gradual increase within large scale manufacturing units, the number of companies to adopt the technology is predicted to rise rapidly according to experts. Major companies such as General Electric, Ford Motors, Nike, Rolls Royce and Mattel use 3D printing in a variety of ways to enhance the production of their products.



The world's largest aviation manufacturer, GE, is able to improve productivity on the most complex designs using 3D printing.

GE Aviation is one of the largest companies to take on the use of 3D printing for its manufacturing process'. A critical component within the development process for its latest LEAP jet engines is the creation of more than 85,000 fuel injection nozzles. Normally these each nozzle is assembled from twenty distinct components but with the

use of 3D printers GE Aviation is able to create each unit in a single print process with no assembly requirements.

The new process has proven to be more efficient than traditional methods on a number of levels, from product strength to per unit cost. It has also allowed engineers to design components that are more complex without the fear of additional manufacturing complexity such as the creation of integrated air passages within the components that allow for easy airflow and cooling which can withstand temperatures of over 2400 degrees Fahrenheit! You can expect to witness these components in action with the use of the next range of Boeing 737 MAX and Airbus A320neo aircrafts, which will be using this next generation engine.

General Electric is also working towards using 3D printing technologies with its range of healthcare and medical devices. Its experiments with ultrasound probing has already proven successful. These small devices are printed with the intent to be used on patients to give a sonar like imaging ability to doctors. Normally these types of probes require a high degree of crafting due to the detailed patterning of their design however with the use of 3D printers they can be created in one seamless step which helps cut costs up to 30 percent. For similar reasons Rolls Royce, a cutting edge engineering